BOARD ACTIVITY SHEET: MARCH 2024

Science and Technology Part - 2

Time: 2 Hours **Max. Marks: 40**

***Note*:** i. *All* questions are compulsory.
ii. The numbers to the right of the questions indicate full marks.
iii. In case of MCQs [Q. No. 1(A)] only the first attempt will be evaluated and will be given credit.
iv. Scientifically and technically correct, labeled diagrams should be drawn wherever necessary.
v. Each new question should be started on the new page.

Q.1. (A) Choose and write correct option number as answer: **[5]**

i. Regeneration occurs in _____.
(A) *Amoeba* (B) *Paramoecium*
(C) *Euglena* (D) *Planaria*

ii. *Lactobacillus brevis* gives us the beverage _____.
(A) cocoa (B) coffee
(C) wine (D) cider

iii. A minor change occurs due to change in position of any nucleotide is termed as _____.
(A) transcription (B) translocation
(C) mutation (D) translation

iv. Atomic power plant in Maharashtra is _____.
(A) Chandrapur (B) Koyana
(C) Tarapur (D) Anjanwel

v. Example of phylum Arthropoda is _____.
(A) scorpion (B) starfish
(C) earthworm (D) *Hydra*

(B) Solve the following: **[5]**

i. Pick the odd man out :
duckbill platypus, pomfret, lungfish, *Peripatus*.

ii. Write co-relation :
Skin: Melanin :: Pancreas : _____

iii. State, true or false :
Production of various useful aquatic organisms with the help of water is called blue revolution.

iv. Write full form of WHO.

v. Match the pair:

	'A'		'B'
(1)	Male	(a)	44 + XX
		(b)	44 + XY
		(c)	44 + YY

Q.2. (A) Give scientific reasons (any *two*): **[4]**

i. Though tortoise lives on land as well as in water, it cannot be included in class-Amphibia.

ii. Hydroelectric energy, solar energy and wind energy are called renewable energies.

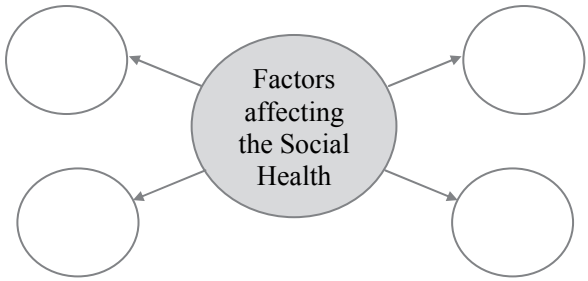
iii. We feel tired after exercise.

(B) Answer any *three* of the following: **[6]**

i. Write *four* adverse effects of radiations on human body.

Target Publications

ii. Redraw and complete the given chart.



iii. Write any *four* important facts which must be thought over while considering the scope of disaster.

iv. Name *two* fuels obtained by microbial processes. Why is it necessary to increase the use of such fuels?

v. What is DNA fingerprinting? Where is it mainly useful?

Q.3. Answer any *five*: **[15]**

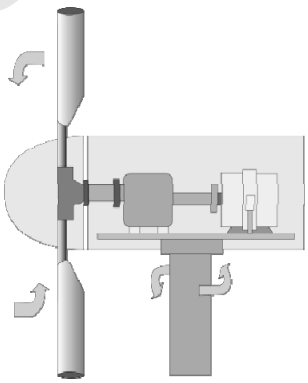
i. (a) What are vitamins?
(b) Classify them according to their solubility.
(c) Give one example of each type as per above classification.

ii. Explain the following terms:
(a) Stem cells
(b) Cloning
(c) Genetically modified crops.

iii. Complete the given paragraph by filling the blank spaces from the options given in the bracket.
[oxygen, pyridines, mechanical, CO_2, petroleum, fatal, polyester, *Nocardia*]

Spilling of _____ oil occurs in ocean due to various reasons. This oil may prove _____ and toxic to aquatic organisms. It is not easy to remove the oil layer from surface of water by _____ method.
However, bacteria like *Pseudomonas* spp. and *Alcanovorax borkumensis* have to ability to destroy the _____ and other chemicals. Hence, these bacteria are used to clear the oil spills. These are called as hydrocarbonoclastic bacteria (HCB). HCB decompose the hydrocarbons and bring about the reaction of carbon with _____, _____ and water is formed in this process.

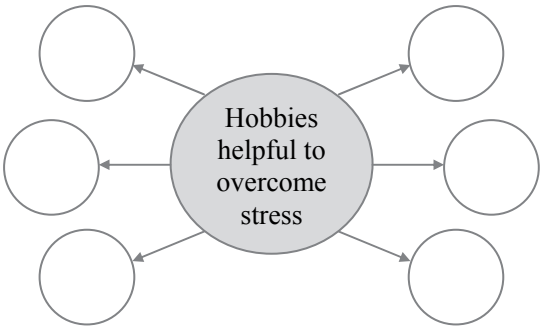
iv. Observe the given picture and answer the questions.



Questions:
(a) Identify the given diagram.
(b) What is the source of energy in this project?
(c) Why this source is supposed to be ecofriendly?

Target Publications

v. Complete the following chart:



vi. Write names of any *six* materials which are to be in the first aid box.

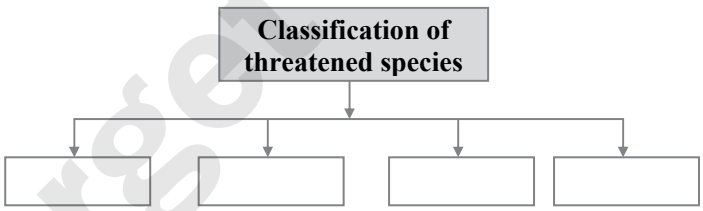
vii. (a) Define vestigial organs.
(b) Write names of any *two* vestigial organs in the human body.
(c) Write name of those animals in which these vestigial organs are functional.

viii. Read the statements and answer accordingly:
(a) Calcareous spines are present on the body. Tube feet are used for locomotion and capturing prey also.
– Identify the phylum and give *one* example.
(b) I live in your small intestine. Pseudocoelom is present in my thread like body.
– Identify the phylum and give *one* example.
(c) My body is spindle shaped. I respire with the help of gills.
– Identify my class and give *one* example.

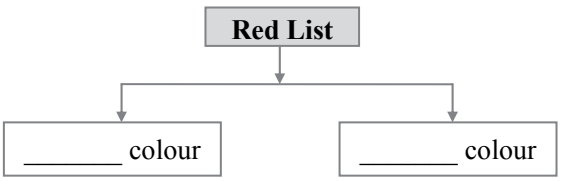
Q.4. Answer any *one*: **[5]**

i. Answer the following questions based on human male reproductive system:
(a) In which organ sperms are formed?
(b) What is the length of a sperm?
(c) What is the work of vas deferens?
(d) By which type of cell division, sperms are formed?
(e) Name any *two* unpaired organs of male reproductive system.

ii. Answer the following questions based on environmental conservation and biodiversity:
(a) Complete the flow chart: 2



(b) How can biodiversity be conserved? Give any *four* points. 2
(c) Complete the flow chart: 1
Colour of the pages in Red List of Endangered Species



Target Publications